

TASK3:

```
vennela_07@VennelaPC:~$ mkdir week3
mkdir: cannot create directory 'week3': File exists
vennela_07@VennelaPC:~$ cd week3
vennela_07@VennelaPC:~/week3$ pwd
/home/vennela_07/week3
vennela_07@VennelaPC:~/week3$ nano task3.c
vennela_07@VennelaPC:~/week3$ gcc task3.c -o task3
vennela_07@VennelaPC:~/week3$ ls
File1.txt  process.c  task2      task3.c      task4.c
File2.txt  task1      task2.c    task3.txt    task5
file1.txt  task1.c    task3      task4        task5.c
vennela_07@VennelaPC:~/week3$ ./task3
Parent Process Started
Parent PID: 882
Parent PPID: 588

Parent Process
Child PID: 883
Parent is Running...

Child Process Started
Child PID: 883
Child PPID: 882
Child is Running...
Child Process Terminated
Child has finished.
Parent Process Terminated
```

```
GNU nano 7.2 task3.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());
    printf("Parent PPID: %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        // Child process
        printf("\nChild Process Started\n");
        printf("Child PID: %d\n", getpid());
        printf("Child PPID: %d\n", getppid());

        printf("Child is Running...\n");

        sleep(5);

        printf("Child Process Terminated\n");
    }
    else
    {
        // Parent process
        printf("\nParent Process\n");
        printf("Child PID: %d\n", pid);
        printf("Parent is Running...\n");

        wait(NULL);

        printf("Child has finished.\n");
        printf("Parent Process Terminated\n");
    }

    return 0;
}
```

TASK4:

```
vennela_07@VennelaPC:~/week3$ nano practicaltask4.c
vennela_07@VennelaPC:~/week3$ gcc practicaltask4.c -o practicaltask4
vennela_07@VennelaPC:~/week3$ ./practicaltask4
Process started
PID: 934
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
Process is Running...
^C
vennela_07@VennelaPC:~/week3$ ps -p 2500 -o pid,ppid,state,cmd
PID    PPID S  CMD
vennela_07@VennelaPC:~/week3$ top
top - 04:00:18 up 37 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7767.6 total,  7097.9 free,   515.1 used,   309.2 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  7252.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21720	13072	9732	S	0.0	0.2	0:00.77	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	66828	15920	14704	S	0.0	0.2	0:00.44	systemd-journal
101	root	20	0	25404	6836	5168	S	0.0	0.1	0:00.46	systemd-udev
111	systemd+	20	0	21336	13132	11008	S	0.0	0.2	0:00.06	systemd-resolve
122	systemd+	20	0	91028	7964	6992	S	0.0	0.1	0:00.10	systemd-timesyn
165	root	20	0	4244	2676	2420	S	0.0	0.0	0:00.02	cron
166	message+	20	0	9644	5380	4672	S	0.0	0.1	0:00.09	dbus-daemon
195	root	20	0	17972	8884	7836	S	0.0	0.1	0:00.07	systemd-logind
200	root	20	0	1756104	10996	8692	S	0.0	0.1	0:00.21	wsl-pro-service
225	syslog	20	0	222516	6052	4628	S	0.0	0.1	0:00.06	rsyslogd
244	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
259	root	20	0	107024	23200	13740	S	0.0	0.3	0:00.09	unattended-upgr
344	root	20	0	6696	4668	3888	S	0.0	0.1	0:00.00	login
399	vennela+	20	0	20108	11372	9448	S	0.0	0.1	0:00.08	systemd
400	vennela+	20	0	21156	3596	1872	S	0.0	0.0	0:00.00	(sd-pam)
416	vennela+	20	0	6080	5384	3708	S	0.0	0.1	0:00.04	bash
584	root	20	0	3188	1116	980	S	0.0	0.0	0:00.00	SessionLeader
586	root	20	0	3204	1256	1108	S	0.0	0.0	0:00.35	Relay(588)
588	vennela+	20	0	6080	5400	3660	S	0.0	0.1	0:00.04	bash
942	vennela+	20	0	9400	5744	3508	R	0.0	0.1	0:00.00	top

```
GNU nano 7.2 practicaltask4.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Process started\n");
    printf("PID: %d\n", getpid());

    while (1)
    {
        printf("Process is Running...\n");
        sleep(2);
    }

    return 0;
}
```